Experiment 4: 7-Segment Display using AT89C51 using Proteus

Aim:

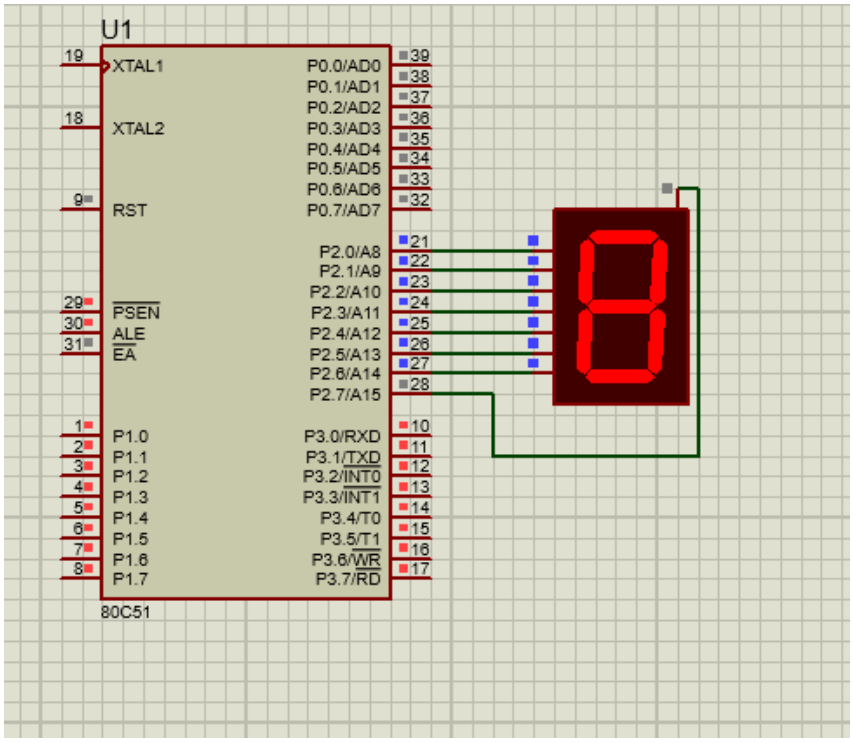
To interface and display numbers on a 7-segment display using AT89C51 in Proteus.

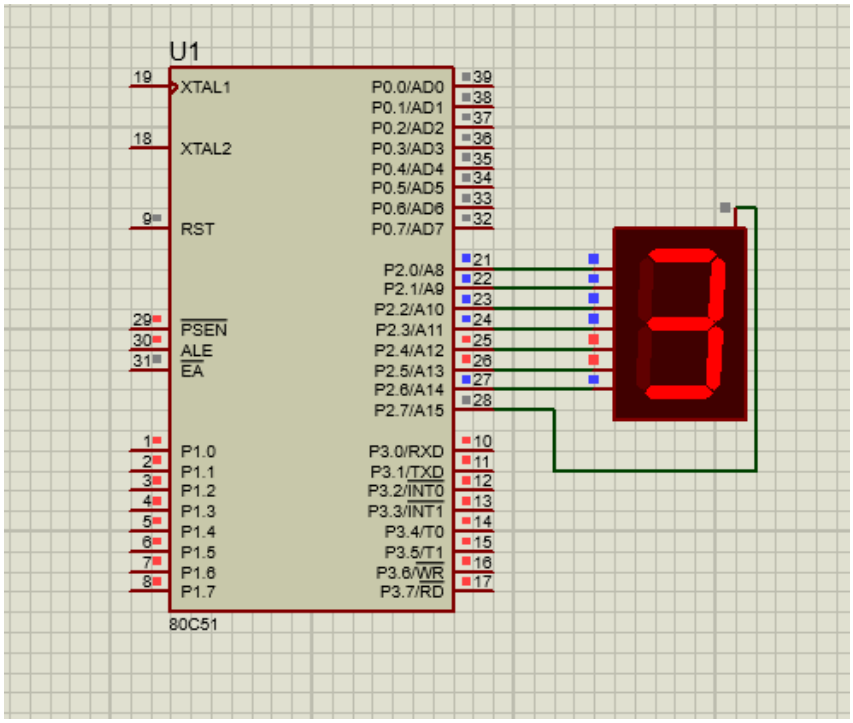
Procedure:

1. Open Proteus and create a new circuit.
2. Place AT89C51, 7-segment display and resistors.
3. Connect the display to the microcontroller port.
4. Write and compile the display program.
5. Load the HEX file into AT89C51.
6. Run the simulation and observe the displayed numbers.

*main.asm

```
ORG 0000H
UP: MOV P2,#0C0H
    ACALL DELAY
    MOV P2,#0F9H
    ACALL DELAY
    MOV P2,#0A4H
    ACALL DELAY
    MOV P2,#0B0H
    ACALL DELAY
    MOV P2,#099H
    ACALL DELAY
    MOV P2,#092H
    ACALL DELAY
    MOV P2,#082H*
    ACALL DELAY
    MOV P2,#0F8H
    ACALL DELAY
    MOV P2,#080H
    ACALL DELAY
    MOV P2,#090H
    ACALL DELAY
    SJMP UP
DELAY:
    MOV R5,#10
H1: MOV R4,#180
H2: MOV R3,#255
H3: DJNZ R3,H3
    DJNZ R4,H2
    DJNZ R5,H1
    RET
```





Result:

The numbers were successfully displayed on the 7-segment display using AT89C51.